A Computational Framework to Predict Direction of Muscle Fiber in Human Muscle Using Stokes Flow

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Abstract

Biophysical simulations of tissue mechanics in realistic geometries can serve as a powerful computational tool for uncovering relationships between form and function. Specifically, skeletal muscle tissue plays a central role in movement and force generation but presents a modeling challenge due to its complex multiscale organization. Muscle geometry and myofiber orientation are known to be correlated across length scales, but their exact relationship remains incompletely understood. Advances in magnetic resonance imaging (MRI), together with improved segmentation and surface reconstruction methods, have made realistic muscle geometries increasingly accessible for computational modeling. However, the internal myofiber architecture required for mechanical simulation remains difficult to measure directly, presenting a challenge for biomechanical simulations where fiber orientation is a key determinant of predicted tissue behavior. While diffusion tensor imaging (DTI) can provide in vivo estimates of fiber orientation fields, its high cost and limited resolution and availability motivate the development of computational alternatives. In this work, we present a reproducible pipeline for generating and analyzing fiber orientation fields from surface geometry using a fluid flow analogy. The workflow converts surface reconstructions into tetrahedral meshes and solves the Stokes flow equations within the domain. The proximal and distal aponeuroses are assigned as the inlet and outlet boundaries, and the resulting velocity streamlines serve as the predicted fiber orientation field. The framework enables quantitative comparison between fiber fields and provides a practical method to test how modeling assumptions affect fiber direction. By making these relations explicit, this pipeline establishes a foundation for studying how uncertainty in fiber organization propagates into biomechanical predictions.

Motivation and Study Design

- Muscle fiber orientation strongly influences predicted stress and strain fields.
- Internal fiber architecture is hard to measure directly in vivo.
- A geometry-driven computational alternative is evaluated across diverse muscle types.

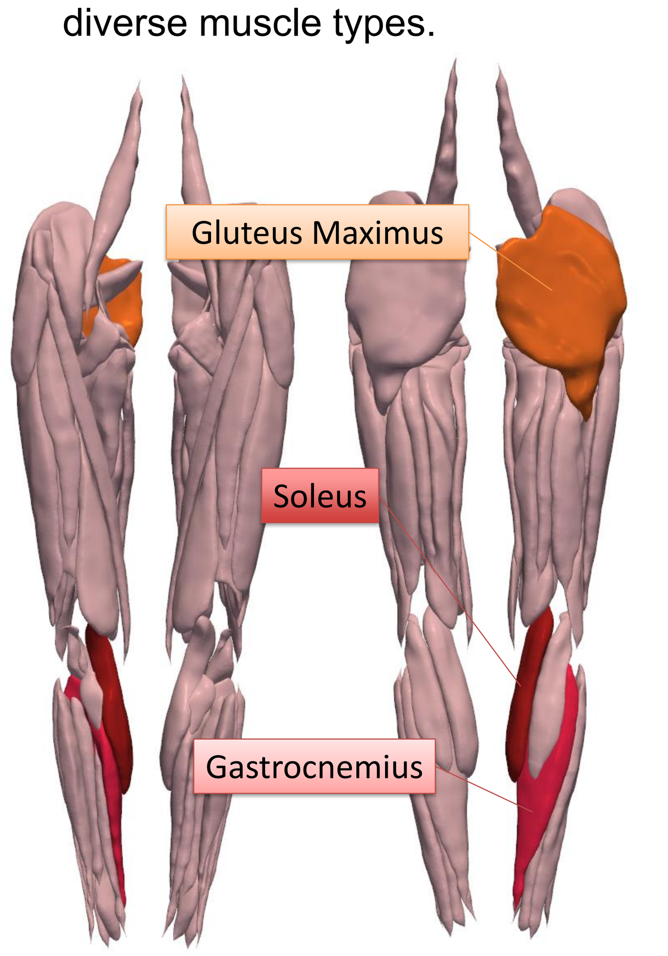

Figure 1. Three anatomically distinct muscles were selected to evaluate the Stokes-flow pipeline: the medial gastrocnemius (bipennate structure), the soleus (complex aponeurosis and pennate architecture), and the gluteus maximus (broad, laterally extended geometry). These muscles provide a range of geometric conditions for assessing fiber prediction and boundary condition sensitivity.

Computational Pipeline for Fiber Orientation

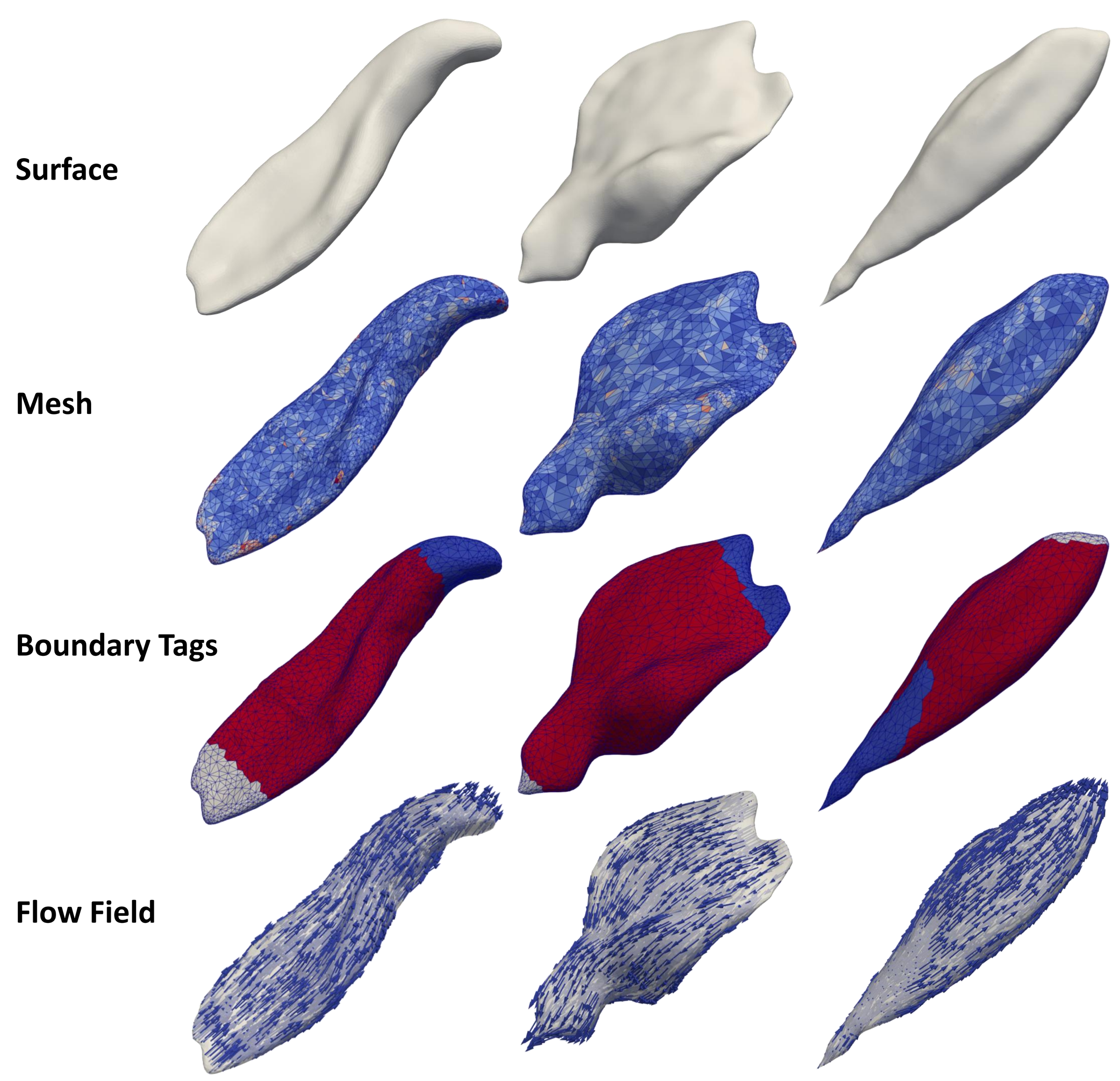

Figure 2. Application of the Stokes-flow pipeline across diverse muscle geometries. Surface reconstructions are first converted to tetrahedral meshes, boundary regions are individually tagged, and Stokes flow is solved to generate fiber orientation fields. Results are shown for the medial gastrocnemius (left), gluteus maximus (middle), and soleus (right).

Boundary Condition Sensitivity

- Variation of inlet traction region and centroid direction alters global streamline alignment and local fiber orientation patterns.
- Angular deviation metrics quantify how inlet boundary selection propagates into spatial differences in predicted fiber fields.

$$\theta_i = cos^{-1}\left(\left|\hat{f}_i^{(1)} \cdot \hat{f}_i^{(2)}\right|\right)$$

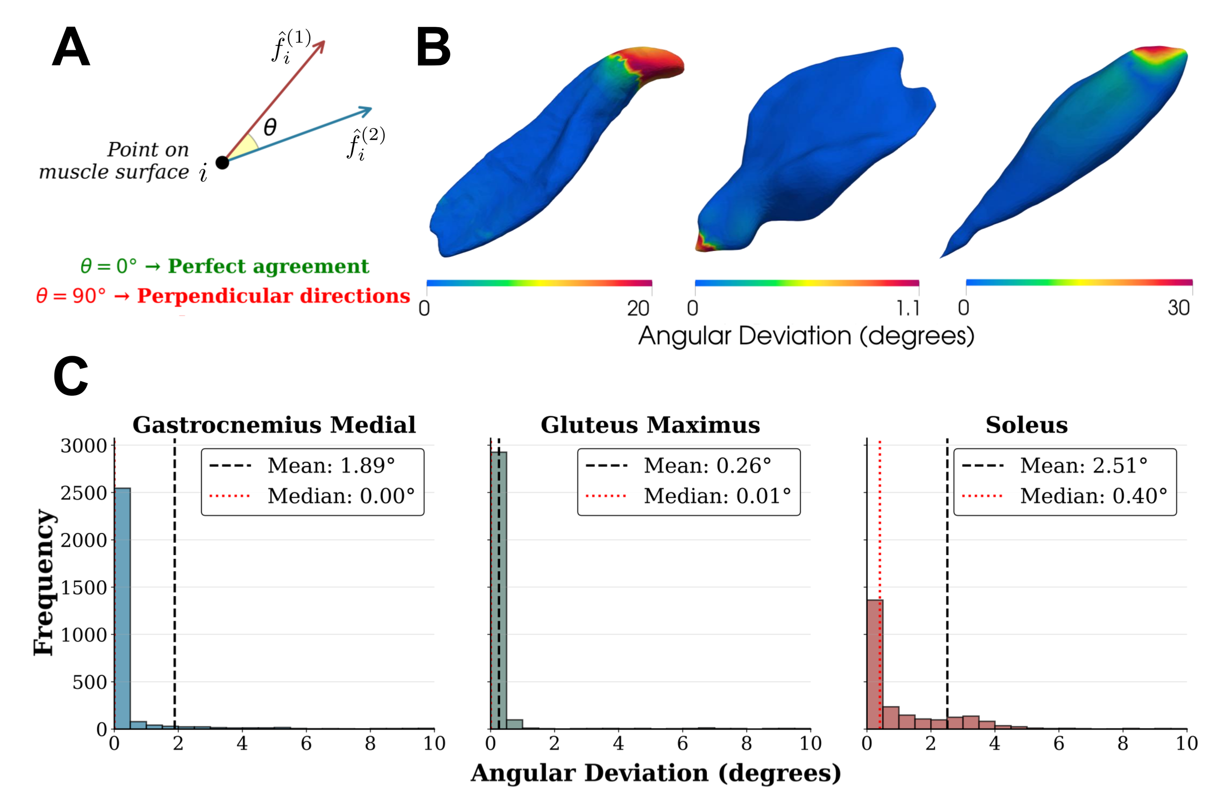

Figure 3. Boundary condition method comparison across muscle geometries. **A)** Angular deviation (θ) between fiber directions at a point on the muscle surface. **B)** Surface heat maps showing angular deviation (degrees) between normal and centroid boundary condition methods for three muscles with distinct geometries: gastrocnemius medial (cylindrical), soleus (pennate), and gluteus maximus (wide/flat). Warm colors indicate higher angular deviations, with differences primarily localized near inlet boundaries. **C)** Distribution histograms reveal highly concentrated, narrow peaks with mean deviations ≤ 2.51° for all three muscles. The minimal deviations across muscle surfaces show that both boundary condition methods produce consistent fiber direction predictions regardless of geometry.

Geometric Sensitivity and Robustness Analysis

- Inlet/outlet regions were dilated and eroded to simulate boundary segmentation uncertainty.
- Predicted fiber orientations show low mean angular deviation from baseline across perturbations.

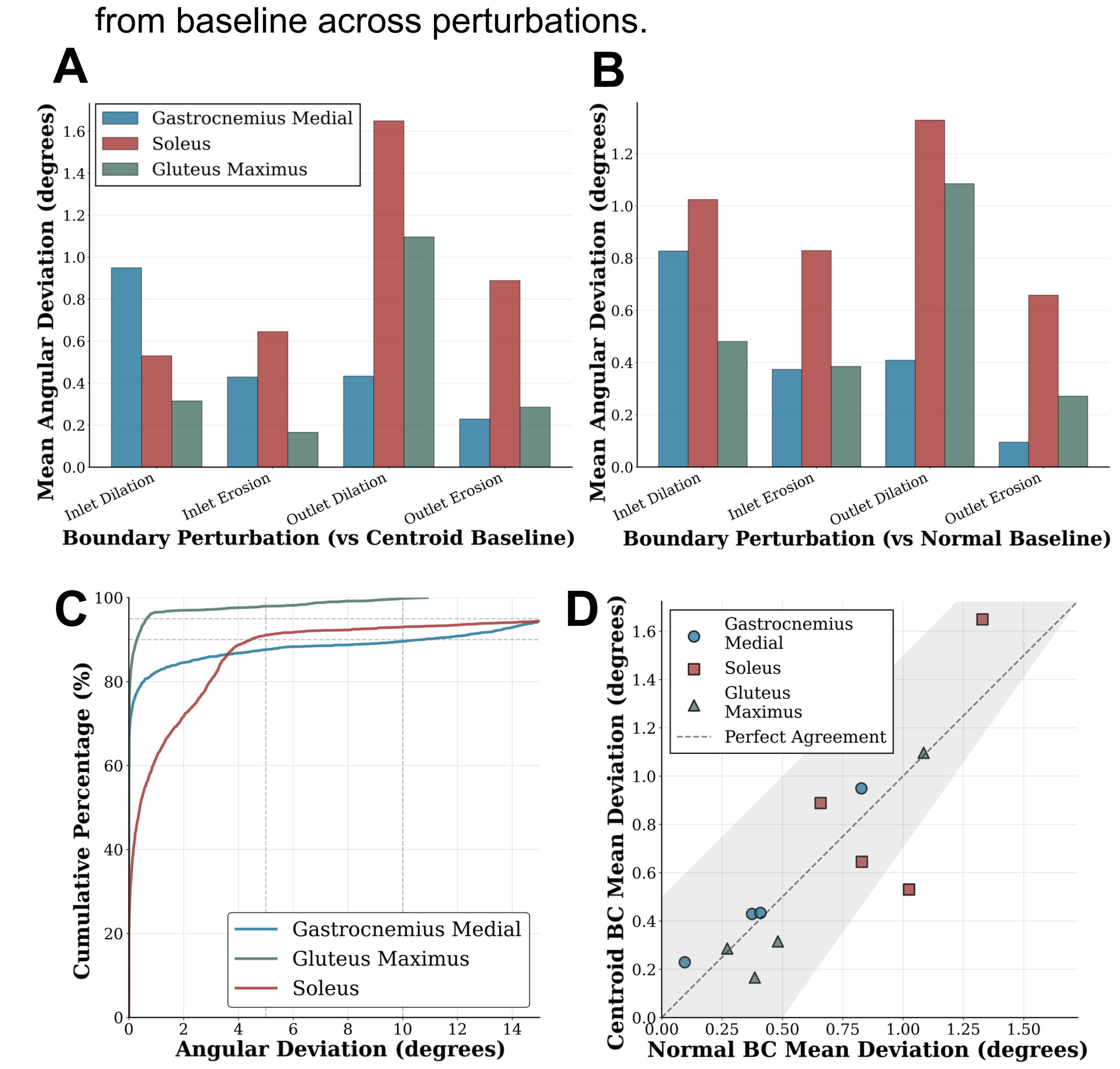

Figure 4. Sensitivity analysis of fiber direction predictions to boundary perturbations. **A-B)** Mean angular deviations for inlet and outlet boundary perturbations (dilation and erosion) relative to centroid baseline **(A)** and normal baseline **(B)** across three muscle geometries. All perturbations produce mean deviations ≤ 1.65°, demonstrating robust fiber predictions. **C)** Cumulative distribution functions showing that ≥88% of surface points exhibit deviations ≤ 5° for all three muscles. **D)** Scatter plot comparing normal and centroid BC methods across all perturbation types, with points clustering near the diagonal line indicating strong agreement between methods regardless of muscle geometry.

Continuum Formulation of the Stokes Flow Model

Steady incompressible Stokes flow is solved within each muscle domain to generate a smooth interior vector field.

$$\nabla \cdot \mathbf{u} = 0, \qquad -\nabla p + \mu \nabla^2 \mathbf{u} = \mathbf{0} \quad \text{in } \Omega.$$

The Cauchy stress tensor defines the viscous stresses and pressure contributions that govern boundary tractions.

$$\boldsymbol{\sigma}(\mathbf{u}, p) = 2\mu\boldsymbol{\varepsilon}(\mathbf{u}) - p\mathbf{I}, \qquad \boldsymbol{\varepsilon}(\mathbf{u}) = \tfrac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T), \qquad \mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}.$$

In one method, a prescribed normal traction is applied at the inlet.

$$\mathbf{t}_{\text{in}} = t_0 \mathbf{n} \quad \text{on } \Gamma_{\text{in}}.$$

In another method, inlet traction is prescribed in a centroid-to-centroid direction between proximal and distal aponeuroses.

$$\mathbf{t}_{\text{in}} = t_0\, \hat{\mathbf{v}}, \qquad \hat{\mathbf{v}} = \frac{\mathbf{c}_D - \mathbf{c}_P}{\|\mathbf{c}_D - \mathbf{c}_P\|} \quad \text{on } \Gamma_{\text{in}}.$$

Side boundaries enforce no penetration and vanishing tangential stress. A stress-free condition is applied at the outlet.

$$\mathbf{u} \cdot \mathbf{n} = 0, \qquad (\mathbf{I} - \mathbf{n} \otimes \mathbf{n})\, \boldsymbol{\sigma}\mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_{\text{side}}. \qquad \boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_{\text{out}}.$$

The governing equations are reformulated in variational (weak) form, allowing mixed finite element discretization and weak [1] enforcement of boundary constraints through Lagrange multipliers.

$$\underbrace{\int_\Omega 2\mu\boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v})\, d\Omega}_{\text{Viscous}} - \underbrace{\int_\Omega p\, \nabla \cdot \mathbf{v}\, d\Omega}_{\text{Pressure}} - \underbrace{\int_{\Gamma_{\text{side}}} \zeta(\mathbf{v} \cdot \mathbf{n})\, d\Gamma}_{\text{Constraint}} - \underbrace{\int_{\Gamma_{\text{in}}} \mu(\nabla\mathbf{u})^T \mathbf{n} \cdot \mathbf{v}\, d\Gamma}_{\text{Inlet stress}} - \underbrace{\int_{\Gamma_{\text{out}}} \mu(\nabla\mathbf{u})^T \mathbf{n} \cdot \mathbf{v}\, d\Gamma}_{\text{Outlet stress}} = \underbrace{\int_{\Gamma_{\text{in}}} t_{\text{in}}\, \mathbf{n} \cdot \mathbf{v}\, d\Gamma}_{\text{Inlet traction}} + \underbrace{\int_{\Gamma_{\text{out}}} t_{\text{out}}\, \mathbf{n} \cdot \mathbf{v}\, d\Gamma}_{\text{Outlet traction}}$$

Normalized velocity vectors define the predicted local muscle fiber orientation field.

$$\hat{f} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

Conclusions

- Muscle fiber architecture can be inferred from realistic surface geometry using a reproducible Stokes-flow formulation.
- Sensitivity analyses demonstrate that predicted fiber orientations remain stable under variations in inlet definition and boundary perturbation.
- Making the geometry to fiber relationship explicit and quantifiable supports future research of structure–function relationships in skeletal muscle mechanics.

References

1 TE Andreassen et al. *Sci. Data* 10.1 (2023).
2 IS Devold, ME Rognes, and P Rangamani. *bioRxiv* (2025)
3 J Varvik. et al. *Int. J. Numer. Methods Biomed. Eng.* 37.12 (2021)

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